Testing quantum memories on IBM hardware

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1. THE EXPERIMENT

This project asks how well a noise model built from IBM's calibration data predicts the errors of a logical quantum memory, and whether a heavy-hex code improves when its distance grows from 3 to 5.

Fifteen jobs ran on `ibm_fez`, a 156-qubit Heron r2 processor, over five days. Each held X- and Z-memory circuits at 1, 2, 3, 4, 6 and 8 rounds, with 2500 shots per memory circuit. Before each job, a prediction built from that day's calibration was computed and saved. No rate in it was fitted to memory data, and no shot was post-selected or discarded.

The model underestimates the failure rate in both memories, and distance 5 does worse than distance 3 on these placements, as the model predicts. The rest of this write-up looks for the missing error with decoder checks, an error budget by noise channel, an analysis of persistent flag and relay readouts, and a follow-up on `ibm_kingston` with a modified circuit. All results apply to the sampled chip regions and circuits, not to heavy-hex codes in general.

2. CODE AND CIRCUITS

On IBM's heavy-hex lattice each qubit has at most three neighbors. The heavy-hex subsystem code combines surface-code and Bacon–Shor checks [1, 2] and is a $[[d^2, 1, (d-1)^2/2, d]]$ code (data qubits, logical qubits, gauge qubits, distance). In the X/Z convention of Ref. [3], its gauge group is

$$\begin{aligned}\mathcal{G} = \big\langle\, & X_{i,j}X_{i,j+1}, \\ & Z_{i,j}Z_{i+1,j}Z_{i,j+1}Z_{i+1,j+1} \quad (i+j \text{ odd}), \\ & Z_{i,1}Z_{i+1,1}\ (i \text{ odd}), \quad Z_{i,d}Z_{i+1,d}\ (i \text{ even}) \,\big\rangle,\end{aligned} \tag{1}$$

where $i, j = 1, \ldots, d$ label data rows and columns, with $i \le d-1$ wherever $i+1$ appears and $j \le d-1$ wherever $j+1$ appears. The stabilizers are generated by

$$\begin{aligned}\mathcal{S}_X :\ & X_{i,j}X_{i+1,j}X_{i,j+1}X_{i+1,j+1} \quad (i+j \text{ even}), \\ & X_{d,j}X_{d,j+1}\ (j \text{ odd}), \quad X_{1,j}X_{1,j+1}\ (j \text{ even}), \\ \mathcal{S}_Z :\ & \textstyle\prod_{j=1}^{d} Z_{i,j}Z_{i+1,j}, \quad i = 1, \ldots, d-1,\end{aligned} \tag{2}$$

and the logical operators are $X_L = \prod_i X_{i,1}$ and $Z_L = \prod_j Z_{1,j}$. Each X stabilizer is one X gauge or a product of two. Each Z stabilizer is the product of the Z gauges between two adjacent rows; it has weight $2d$, and this Bacon–Shor half has no threshold [1, 4].

In X memory all data qubits start in $|+\rangle$ and are read in X after n rounds of gauge measurements; a shot fails if the decoded X_L differs from its initial value. In Z memory the data qubits start in $|0\rangle$ and are read in Z. X memory fails through Z errors and Z memory through X errors.

Measuring the gauges. Each round measures all gauges of the memory basis, then those of the other basis. An X gauge needs one ancilla and two CX gates. A bulk Z gauge acts on four data qubits that its ancilla cannot reach directly, so two neighboring X ancillas act as flags: they collect the data parities, pass them to the Z ancilla and are uncomputed (Fig. 2a). An edge Z gauge's ancilla is coupled to neither of its two data qubits and reaches each through its own relay. Without faults, flags and relays read 0. Counting data qubits, X ancillas, Z ancillas and relays, the distance-3 patch uses $9+6+4+4 = 23$ physical qubits and distance 5 uses $25+20+12+8 = 65$. The twelve-CX bulk gadget has four more CX gates than the flag circuits of Refs. [1, 3]. Any single fault leaves at most one data error, or two in the same column, up to the measured gauge, so dropping the flag outcomes does not reduce the X-memory circuit distance.

Detectors and decoding. Let $m_{s,t}$ be the value of stabilizer s in round t, the XOR of its gauge outcomes. The detector

$$D_{s,t} = m_{s,t} \oplus m_{s,t-1} \tag{3}$$

fires when this value changes. Memory-basis stabilizers give one detector per round and one at the final data readout; the first compares with the value fixed by preparation. Z memory also uses the flag and relay readouts as detectors. Before the first round the other-basis gauges are measured once, so every round runs the same gate sequence; those outcomes enter no detector.

Stim builds the detector error model and PyMatching decodes it by minimum-weight perfect matching [5–7]. Unless stated otherwise, device and simulated shots are decoded with the priors of the stored prediction. The circuit distance, the fewest undetected faults that flip the logical value, is d in both bases according to exhaustive low-weight and Stim fault searches, although the distance-5 Z matching graph has a shorter logical path (Sec. 5).

Timing and placement. The circuits use CZ, $\sqrt{X}$, X and virtual R_Z gates with explicit waits. Waits of 500 ns or more after a qubit's first gate get two X pulses, at one quarter and three quarters of the wait; IBM's own decoupling and twirling were off. A round takes about 7.9 µs. Readout and reset take 6.7 µs of it on the critical path, so the data qubits sit idle for 94–97% of the round (Fig. 2b).

Placements were ranked by an estimated error per

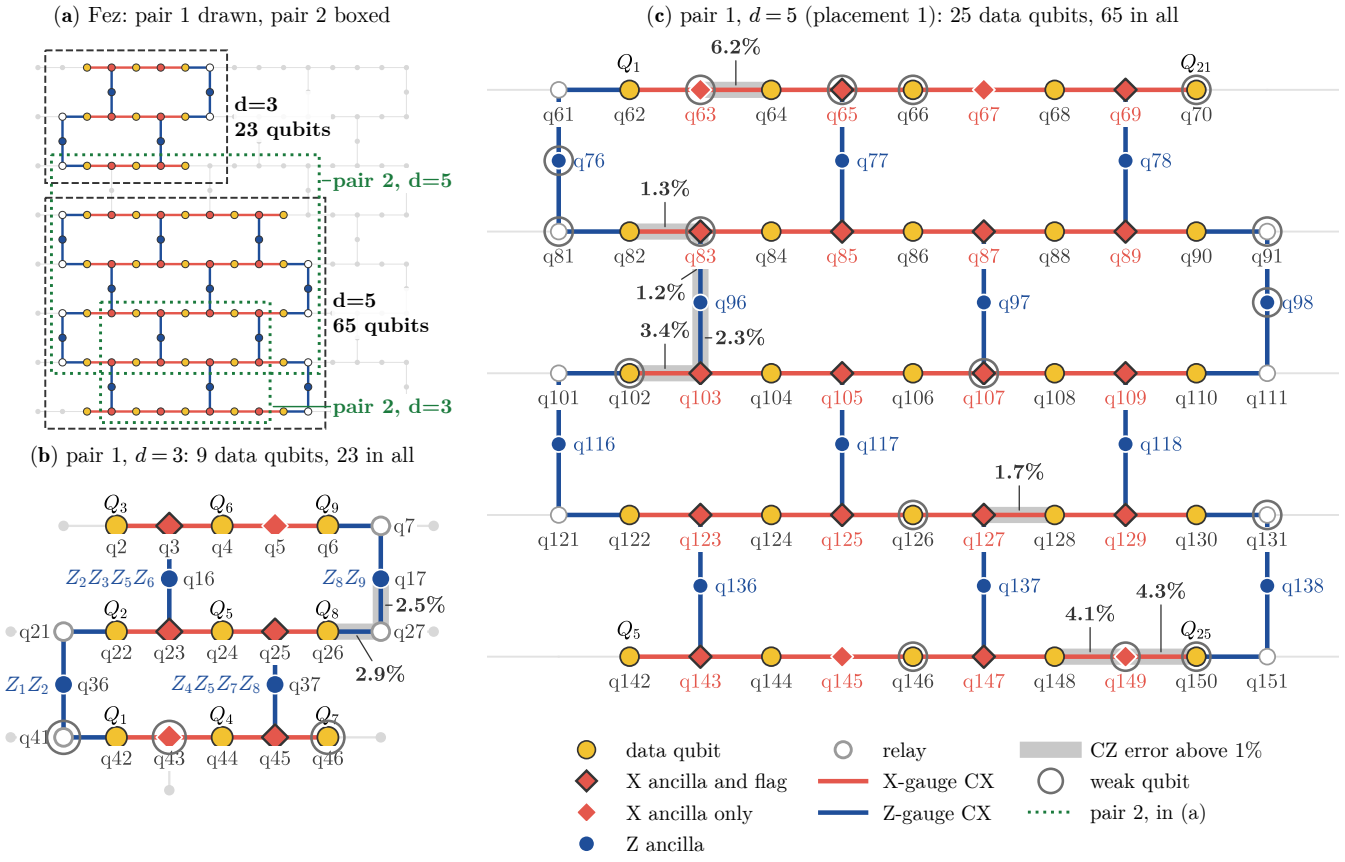


FIG. 1. Fez layouts. (a) Pair 1 (black boxes) and pair 2 (green boxes, separate jobs). (b,c) Pair 1's distance-3 and distance-5 patches; Q labels are data qubits and q labels are physical qubits. Shaded couplers have CZ error above 1%; rings mark readout error above 5%, $T_1 < 50\,\mu$s or $T_2 < 30\,\mu$s. The large patch contains weak parts, including q149 and q150 with a reported T_2 of only 5–6 µs.

round,

$$E = \sum_{\text{CZ}} r_{qq'} + \sum_{\text{meas., reset}} e_q + \sum_{\text{data}} \frac{9\,\mu\text{s}}{2T_{1,q}}, \qquad (4)$$

with gate and readout errors $r_{qq'}$ and e_q, and 9 µs a pre-compilation estimate of the round time. A placement is usable if no qubit or coupler in it is marked faulty or has a reported error of 50% or more. Every day had eighteen usable distance-3 placements but only two usable distance-5 placements, the same two each day, sharing 51 of their 65 qubits. A 5% CZ-error cutoff or a 10% readout-error cutoff would have left no distance-5 placement at all. The selected large patches include CZ errors of 3.4–7.0% and readout errors of 8–14%, and one relay reached 48.6% on day 4.

Each day, each distance-5 placement was paired with the best distance-3 placement not yet used that day, preferably one not touching it. Pair 1 ran both patches in one job, on disjoint qubits that stayed the same every day. No free distance-3 placement fit beside pair 2's large patch, so its small patch ran in a separate job about a minute later and overlapped it. One pair on one day is a *pair-day*: ten pair-days, two memories and two distances give forty error curves. The pairs share qubits, and the large patches reuse theirs every day, so the pair-days are not independent samples.

3. NOISE MODEL AND IDLE TESTS

The model follows the submitted circuits gate by gate and wait by wait. IBM's average gate infidelities become depolarizing channels,

$$p_1 = \tfrac{3}{2}\, r_q, \qquad p_2 = \tfrac{5}{4}\, r_{qq'}, \qquad (5)$$

spread equally over the 3 or 15 nonidentity Paulis [9]. Readout and reset both use the reported readout error e_q. A wait t gets Pauli-twirled relaxation and dephasing [10],

$$\begin{aligned} p_x = p_y &= \tfrac{1}{4}\big(1 - e^{-t/T_1}\big), \\ p_z &= \tfrac{1}{2}\big(1 - e^{-t/T_2}\big) - \tfrac{1}{4}\big(1 - e^{-t/T_1}\big), \end{aligned} \qquad (6)$$

with IBM's T_2 capped at $2T_1$. During decoupled waits T_2 becomes T_2^{DD}, with a dephasing multiplier f,

$$\frac{1}{T_2^{\text{DD}}} = \frac{1}{2T_1} + f\,\Gamma_\phi, \qquad \Gamma_\phi = \frac{1}{T_2} - \frac{1}{2T_1}, \qquad (7)$$

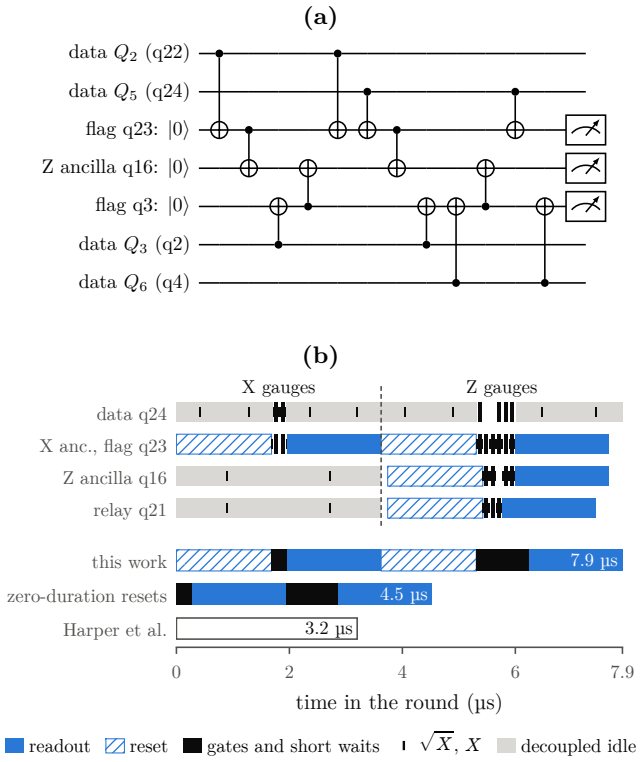


FIG. 2. (a) A bulk Z-gauge measurement through two flags and the Z ancilla, with twelve CX gates, each compiled as H·CZ·H. Without faults the flags return to 0. (b) One compiled Fez round, one qubit per role. Readout and reset dominate the 7.9 µs round; the bars below show the same round with zero-duration resets and the parallel, reset-free circuit of Ref. [8].

so $f = 0$ leaves only T_1 decay and $f = 1$ gives IBM's echo dephasing. Waits under 500 ns get no pulses and use the capped T_2. A qubit ideally in $|0\rangle$ gets no idle channel until its next gate, which ignores relaxation after a reset error.

The model has no leakage, coherent errors, ZZ crosstalk, drift within a job or state-preparation errors. Isolated benchmarks may also miss errors that appear only in parallel operation or in mid-circuit readout and reset, while the single-qubit gates around each CZ may count error that its benchmark already includes, so the bias can go either way.

Before each job, predictions at $f = 0$ and $f = 1$ were simulated with 200,000 shots per circuit and saved; the one at $f = 1$, the *stored prediction*, is the main reference. Each job also held ten idle circuits, with the memory schedule but every CZ replaced by an equal wait, read in X and Y after 1, 2, 4, 8 and 16 rounds. The contrast

$$v(n) = \sqrt{\langle X\rangle^2 + \langle Y\rangle^2} \tag{8}$$

is insensitive to frequency offsets, and each data qubit's f comes from

$$\ln v(n) = \alpha + \ln \bar{X}_0(n) - f\big[\ln \bar{X}_0(n) - \ln \bar{X}_1(n)\big], \tag{9}$$

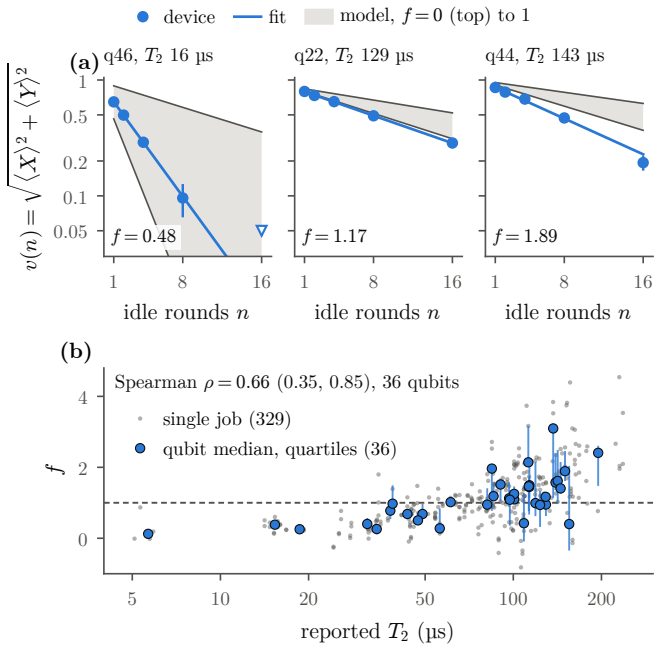


FIG. 3. Idle decay. (a) Three data qubits of day 1's distance-3 pair-1 patch, with the $f = 0$ and $f = 1$ model curves and the fit; the open point is below the 0.05 contrast cutoff. (b) Fitted f against IBM's reported T_2, with per-qubit medians and quartiles.

where $\bar{X}_0$ and $\bar{X}_1$ are the exact model curves at $f = 0$ and $f = 1$ and α absorbs a contrast offset. The median f over data qubits gives a second, *in-job* prediction. This test detects the Z and Y errors behind X-memory failures but not the X flips behind Z-memory failures, and its ancillas stay near $|0\rangle$, unlike those of a memory circuit.

Qubits do not share one f. Forcing one f on all data qubits gives $\chi^2/\nu = 12$–80 in every job, and f grows with the reported T_2 (Spearman $\rho = 0.66$ over 36 qubits, Fig. 3b). For q150, with a reported T_2 of 5–6 µs, the job's median f predicts a Z/Y error of 32–39% per round, while that qubit's own fit gives 4–13% on four days. Per-qubit values of f change the comparison between placements but do not close the device–model gap. Let $\Gamma_{\mathrm{DD}} = f\,\Gamma_\phi$ be the measured dephasing rate beyond T_1 during decoupled waits. Its medians per physical qubit follow

$$\Gamma_{\mathrm{DD}} = 0.11\,\Gamma_\phi + 0.0071\,\mu\mathrm{s}^{-1}, \tag{10}$$

whose constant term is a dephasing floor: a phase flip of about 2.5% over a round's 7.1 µs of decoupled idle, or 1.6–1.8% with other qubit selections or weights. Both sides of this fit contain IBM's Γ_ϕ, so the fit describes the data without isolating a physical channel.

Calibration checks. The calibration snapshots were 21–88 minutes old at the first job, but many values had not changed from earlier snapshots; on day 1 every value matched a snapshot 268 minutes old. The reported readout errors come in steps consistent with 4096 shots per

prepared state, which means about 11% relative shot error at the median readout error.

Using IBM's smaller initialization error for the resets lowers the predicted Z-memory decay by 0.6–0.8 percentage points and mostly worsens the detector agreement; the larger reset and readout errors fitted in Ref. [8] overshoot. No single mid-circuit reset error fixes the discrepancy.

4. FEZ RESULTS

At one to four rounds the device fails more often than the stored prediction in all 160 combinations of day, pair, basis, distance and depth (Fig. 4). Averaging the pairs within each day and then the five days, the shortfall is 3.8–7.6 percentage points in X memory and 8.8–16.6 in Z memory. The 160 cells share qubits and days, so the intervals in Fig. 4 come from the five daily means. Fifteen of the sixteen exclude zero; seven of these lose that exclusion when one day is left out, but every daily mean is positive.

To compare depths, each curve is fitted to

$$1 - 2P_{\mathrm{L}}(n) = A\,(1 - 2\varepsilon)^n \tag{11}$$

by binomial maximum likelihood over all six depths. Here ε is an effective decay per round over the fitted depths and $\Lambda = \varepsilon_3/\varepsilon_5$; $\Lambda > 1$ would mean distance 5 helps. A is free, with $A(1 - 2\varepsilon) \leq 1$. Twelve Z-memory fits have $A > 1$, so A is not a preparation contrast, and saturation near $P_{\mathrm{L}} = 1/2$ leaves little rate information at late depths.

Measured ε exceeds the stored prediction on 39 of 40 curves. The geometric-mean device/model factors are 1.27 and 1.19 in X memory at distances 3 and 5, and 2.06 and 1.84 in Z memory. Other fit windows give 1.15–1.28 in X and 1.65–2.10 in Z, and updated idle inputs do not remove the gap.

Distance and day. Distance 5 fails more often in X memory in 59 of 60 day–pair–depth combinations. With in-job priors, $\Lambda_{\mathrm{X}} = 0.67$ and 0.49 for the two pairs, both below one as predicted, although this comparison mixes distance with placement quality.

For pair 1 the predicted Λ_{X} swings between 0.54 and 0.82 over the five days, while the measured one hardly moves (Fig. 5a). The largest swing is on day 2, when four neighboring distance-3 qubits were reported as much worse than usual but the measured Λ_{X} did not move; setting those qubits to their medians on the other days lowers the predicted Λ_{X} by 0.15.

5. ERROR BUDGET AND DECODER CHECKS

Separating the model's channels. The model splits into CZ gates, single-qubit gates, readout and reset, idle relaxation, and idle dephasing; the idle channel factors into relaxation followed by a phase flip, so the last two can be varied separately. All 31 nonempty combinations were simulated for all fifteen jobs at 20,000 shots per circuit, decoded with the full model's decoder. The channels interact, so switching one off does not remove a fixed share of the logical error. The Shapley value [11] averages each channel's marginal effect over every order of introducing the five channels,

$$\phi_c = \frac{1}{5!} \sum_{\pi} \left[\varepsilon(P_\pi^c \cup \{c\}) - \varepsilon(P_\pi^c)\right], \tag{12}$$

where P_π^c holds the channels before c in the order π; the shares add up to the full model's ε (Table 1).

Idle noise accounts for about half of the modeled decay and the CZ gates for about a quarter. Scaling the dephasing by about 1.5 brings the X-memory decay and detector rates close to the device's but leaves Z memory unchanged. A two-channel fit matches all four decay summaries with relaxation scaled by 2.63 and dephasing by 0.60, yet the detector rates still disagree, so the summaries do not identify the missing channel.

Scaling every error rate of the model by one factor k changes ε faster than linearly, with $d\ln\varepsilon/d\ln k = 1.3$–$1.4$ near $k = 1$, while Λ hardly moves (0.625 to 0.622 for pair 1).

Circuit distance and the matching graph. The distance-5 Z matching graph has a logical path of only three edges; two of them appear because Stim decomposes data X errors that also fire flags and relays. To see whether this matters, the flag and relay detectors were dropped, keeping every gate, readout and the logical observable. The reduced Z graph has distance 3 or 5, and with at most twenty stabilizer detectors at depths 1–4, exact maximum-likelihood (ML) decoding is possible by summing the model probability of every syndrome and logical value. On the device shots, reduced matching changes the mean failure by -0.04 to $+0.24$ percentage points and exact ML by -0.92 to -0.34. The gap to the model stays at 8.5–16.8 points, positive in every pair-day.

For distance-3 X memory at depths 1–4, exact ML changes the device failure by -1.64 to $+1.16$ points and leaves gaps of 2.76–11.06 points. With these priors, matching miscorrects some single faults; at four rounds exact ML also miscorrects 80% of their total probability, so most of these errors are the model's own most likely outcome rather than failures of matching.

TABLE 1. Shapley shares of the in-job prediction's ε, in percent, averaged over the ten pair-days; rounded shares need not sum to 100.

	CZ	1-qubit	Readout/reset	T_1 idle	Dephasing
X, $d = 3$	22	10	12	22	35
X, $d = 5$	25	7	17	17	34
Z, $d = 3$	25	8	13	53	0
Z, $d = 5$	30	9	16	46	0

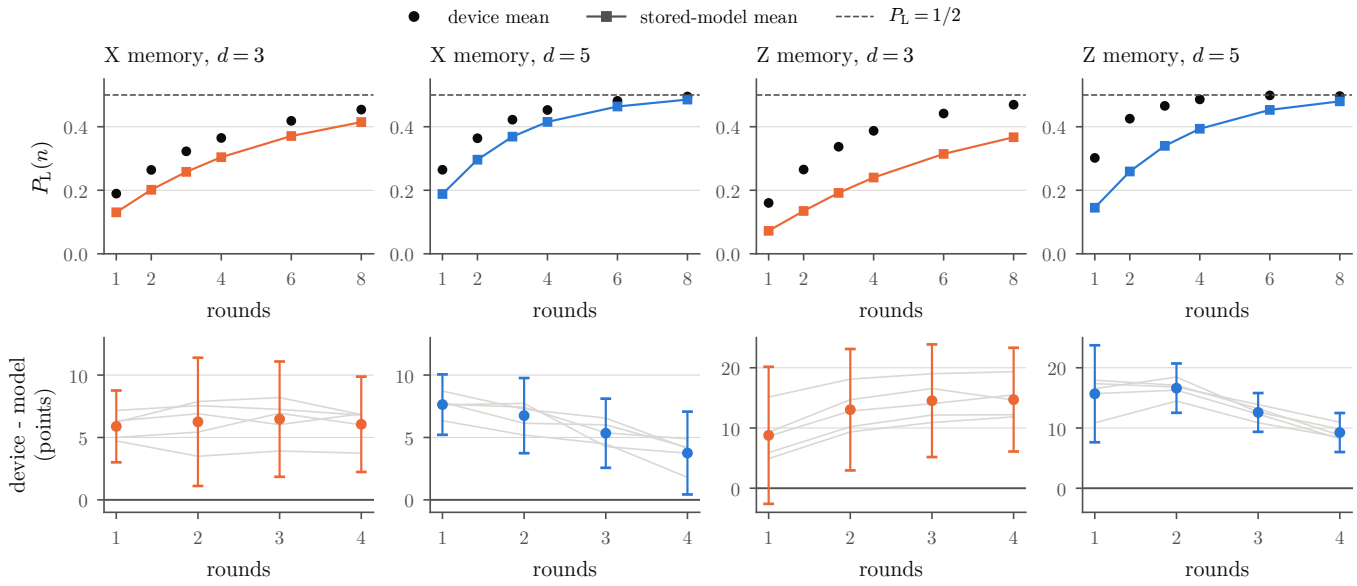


FIG. 4. Fez, decoded with the stored $f = 1$ priors on device and model. Top: failure probability averaged over days and placements; model points are joined, not fitted. Bottom: device minus model at one to four rounds. Gray lines are daily means; bars are approximate simultaneous 95% intervals from the five daily means (t with four degrees of freedom, Bonferroni over sixteen cells).

Priors. Calibrated priors outperform uniform weights. Learning the pair-edge priors from detector correlations [12] changes the X-memory decay only a little, and learning every edge type makes it worse; both are in-sample checks. None of the decoders tried closes the gap.

6. PERSISTENT FLAG AND RELAY READOUTS

On Fez in Z memory, a flag or relay that reads 1 reads 1 again in the next round 23.0% of the time, against 8.2% after a 0; the model gives 6.9% in both cases. For a binary readout x_t define the persistence

$$\Delta(\ell) = \Pr(x_{t+\ell} = 1 \mid x_t = 1) - \Pr(x_{t+\ell} = 1 \mid x_t = 0). \tag{13}$$

Over 320 series per basis the next-round persistence averages 14.8 percentage points in Z and 15.2 in X, against about zero in the model, and it decays with the lag (Fig. 5b). Rates that only differ from shot to shot would give a persistence independent of the lag, so they cannot explain this alone.

A hidden state. A two-state hidden Markov model [13, 14] describes each readout series: a persistent state is entered with probability a per round and left with probability b, and a readout is 1 with probability u_0 outside the state and u_1 inside. The four-, six- and eight-round Z-memory records give medians $a = 2.5\%$, $b = 22.3\%$ and $u_1 = 0.51$. With round time τ the state's lifetime is

$$T_{\mathrm{p}} = -\frac{\tau}{\ln(1-b)} \simeq 31\,\mu\mathrm{s}, \tag{14}$$

and its mean residence time τ/b is about 35 µs. Its occupation starts near 3%, grows toward 10% and averages 6.3% over these circuits; X memory gives similar rates. The state is defined by the readout record, not by a measured transmon level.

Leakage could produce such persistence and spread it through CZ partners. A simple leakage simulation fitted to the detector statistics, however, does not reproduce the logical errors of both memories: every variant leaves a Z-memory deficit, most overshoot the X-memory error, and the results depend strongly on how a leaked qubit acts on its CZ partner.

Streaks and failures. After two consecutive flag or relay 1s, nearby X-memory stabilizers fire 8.4 percentage points more often two rounds later, but they already fire 5.6 points more often two rounds before the streak. Logical failure is 4.8 points higher in X memory, against 0.8 in the model, and at a fixed total of flag and relay 1s the streak association drops to 0.2 points.

7. FOLLOW-UP ON KINGSTON

The follow-up used another Heron r2 processor, `ibm_kingston`, and three circuit changes: no mid-circuit resets, XY4 decoupling instead of paired X pulses, and randomized (twirled) CZ layers. Without a reset, an ancilla starts the next gadget in $X^m|0\rangle$, where m is its last outcome, so each gauge value is the XOR of the current outcome and known earlier ones. A round takes 5.8 µs at distance 3 and 6.0 µs at distance 5. Two memory jobs covered rounds 1–4, with 8192 shots per circuit in total

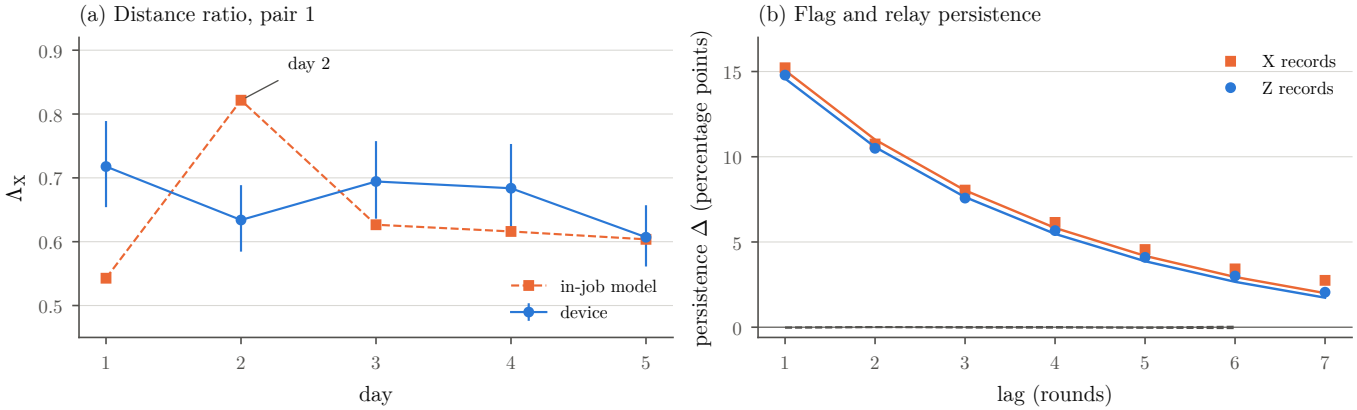


FIG. 5. (a) Pair 1's Λ_X by day, with in-job priors on both sides; device bars are 68% profile intervals. (b) Mean flag and relay persistence with the fitted hidden-Markov curves; the Pauli model's records sit near zero.

and sixteen randomizations per job.

The placement score (4) used the data qubits' T_2 in place of T_1. No distance-5 placement met the cutoffs on T_1, T_2, readout and CZ error, so the lowest-score placement was used; its two worst couplers had CZ errors of 6.4% and 7.1%. The distance-3 patch sat inside the distance-5 footprint, with different roles and schedules. Chip, placement and circuit all changed at once, so the improvement below cannot be credited to any single change.

Fitted over the same depths with the same decoder, the decay per round is 0.40–0.53 of Fez's. The distance ratios stay below one, $\Lambda_X = 0.69 \pm 0.04$ and $\Lambda_Z = 0.57 \pm 0.03$ (fit errors only).

Measured idle inputs. Here the X/Y idle test fits each data qubit's contrast to

$$\ln v(n) = c + \ln G(n) - t_{\mathrm{eq}}(n)/T_q, \tag{15}$$

where T_q is the measured decay time, t_{eq} the time the qubit spends on the equator and G holds the modeled gate, readout and Z-axis relaxation losses. The median measured T_q is 0.74 of IBM's T_2, but four qubits reported at 16–24 µs show 59–196 µs under XY4. Eleven of the twenty-five rates moved by more than three standard errors between the two jobs, 35 minutes apart.

TABLE 2. Kingston: fitted decay, device/model. The stored prediction uses its own priors on both sides; the rows below it share one fixed decoder built from the measured decay.

	X memory		Z memory	
	$d = 3$	$d = 5$	$d = 3$	$d = 5$
Stored prediction	1.90	1.69	1.43	1.12
IBM inputs	1.89	1.53	1.38	1.12
Measured T_q	1.69	1.69	1.25	1.22
Joint idle fit	1.64	1.63	1.14	1.15
Separate X/Y and Z fits	1.69	1.69	1.02	1.08

With measured T_q and a fixed decoder, the pooled shortfalls are 2.7–10.0 percentage points in X memory and 0.2–2.9 in Z, and a few single-job Z points fall below the model. Similar fractional errors at both distances cancel in the distance ratio,

$$\frac{\Lambda^O}{\Lambda^M} = \frac{\varepsilon_3^O/\varepsilon_3^M}{\varepsilon_5^O/\varepsilon_5^M}, \tag{16}$$

with O the device and M the model. The model can therefore get Λ right while the device's X-memory decay is about 69% above the model's.

One idle model does not fit both bases. A Z idle test prepares $|0\rangle$ or $|1\rangle$ and reads Z; the median flip probability is 5.3% per round. Each idle test mainly fixes the combination $t_Z/T_1 + t_{\mathrm{eq}}/T_q$, so the two rates come out strongly anticorrelated (median correlation −0.977). A joint fit with non-negative relaxation and pure dephasing puts seventeen of the twenty-five qubits on a constraint boundary; without the constraints, fourteen need negative pure dephasing, seven of them by more than two standard errors. The constrained fit underestimates the Z-idle decay by 11% and overestimates the X-idle decay by 3%. Fitting each basis separately brings Z memory close to the model (Table 2, last row), but with different effective noise in each basis, so there is no single physical picture. The tests were not interleaved, so drift may also play a part.

Readout calibration. The first calibration listed q72 and q12 at 18–20% readout error; the idle offsets and the calibration 88 minutes later put them near 5%. Re-predicting with the later readout errors and the same decoder leaves gaps of 4.6–11.6 percentage points in X and 3.1–5.5 in Z, so one readout recalibration widens the gaps by 1.1–1.8 points in X and 2.2–3.1 in Z.

Raw readout signals. Ten flag, relay and X-ancilla qubits were put through thirty cycles of H, a 0.41 µs wait, H, readout and reset, with a 4.87 µs period and 3000 shots, and the raw IQ signal was recorded. Besides this reference arm A, arm B added CZs with each qubit's usual partners and arm C dropped the resets. Each arm

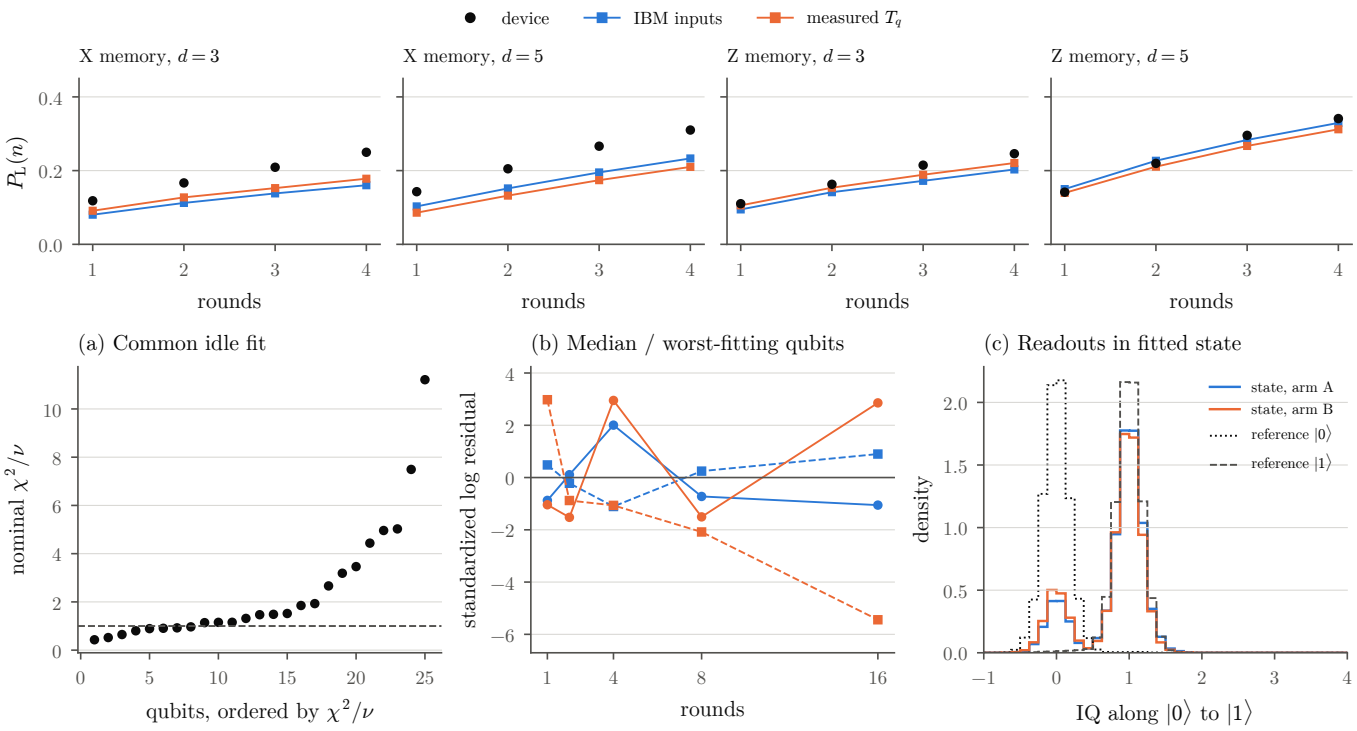


FIG. 6. Kingston. Top: failure probability of device and models, all decoded with one decoder whose priors come from the pooled measured T_q, with edge probabilities capped at 0.1. Bottom: (a) reduced χ^2 of the joint idle fit for each data qubit; (b) standardized log residuals for a median and the worst qubit; (c) raw readout signal in the fitted persistent state, projected on each qubit's 0-to-1 axis. Most of it stays inside the reference clouds; no $|2\rangle$ reference was available.

was a separate job, and arm B ran about eight minutes after arm A.

Only 0.02–0.18% of the readouts fall outside both reference clouds. In the fitted persistent state most signals stay inside the clouds, and about 60% read 1 (Fig. 6c). The state's median lifetime is 40 µs in arm A and 32 µs in arm B, close to Fez's 31 µs. Without a calibrated $|2\rangle$ cloud this does not rule out leakage of the measured qubit, and a similar lifetime does not prove the same cause on both chips.

With CZs, qubits that share a partner enter the state together (median correlation 0.13, against 0.009 for other pairs), and the onset rises by about 0.09% per CZ. Without CZs the state is present from the first readout, and its response to the fixed wait is consistent with a frequency shift of unknown origin. In the reset-free memory, flag readouts still persist, while relay readouts show no excess persistence.

8. TAKEAWAYS

- The calibration-based Pauli model underestimates the decay per round on both chips: by factors of about 1.2–1.3 in X memory and 1.8–2.1 in Z memory on Fez, and about 1.7 in X memory on Kingston even with measured idle decay.
- Distance 5 did worse than distance 3 on both chips, as the model predicted for these circuits and placements. The model's fractional errors are similar at both distances and largely cancel in Λ.
- On Kingston, the circuit without resets, with XY4 decoupling and twirled CZs, had about half of Fez's decay per round, but the chip and placement also changed.
- Flag and relay readouts show a persistent state that the model lacks. On Kingston it mostly reads as an ordinary 0 or 1, and its onset was higher in the arm with CZs, which ran eight minutes later; its source is still open.

CODE AND DATA

The circuits, calibrations, stored predictions, shots and analysis scripts, with reproduction instructions, are at `https://github.com/GoofyGgoober/heavyhex-subsystem-code` (tag `paper`).

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